

Trevor Henry Chiboora

Security Engineer | SOC & EDR | Penetration Testing | Detection Engineering

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Based in Newark, NJ. Relocating to Kigali, Rwanda, available January 2027. Open to remote roles.

Summary

Security engineer with about three years in security operations, penetration testing and security research at the Upanzi Network, Carnegie Mellon University Africa (Kigali), and Approov (UK). Before that, five years in IT operations and teaching. I have run full-cycle penetration tests, from testing through retest; monitored 15+ servers with SIEM, IDS and EDR tooling; and operated internet-facing honeypots for a year. First author of an ACM paper on mobile banking security. Individual AfricaCERT award recipient. Completing an MS in Cybersecurity & Privacy at NJIT (December 2026).

Skills

SOC, EDR and IR: alert triage, threat hunting, escalation, EDR host isolation, runbooks; Elastic Security, Elastic Defend, Wazuh, Security Onion, Suricata, MISP

Penetration testing: web apps, APIs and Linux servers; Burp Suite, OWASP ZAP, Nmap, Masscan, OpenVAS, Nuclei, Nikto, Gobuster, sqlmap, Metasploit, Kali

Mobile and malware analysis: OWASP MASVS, MobSF, jadx, apktool, Frida, Objection; CyberChef, Ghidra, VirusTotal, ANY.RUN; honeypots (Cowrie, OpenCanary)

Infrastructure and access: firewall rules, VPN, network segmentation, backups and recovery, Linux hardening, Active Directory, Windows Server, SSH key and sudo management, access reviews

Detection and AI security: Wazuh rule writing, MITRE ATT&CK mapping, prompt-injection testing, OWASP Top 10 for LLM Applications

Scripting: Python, Bash, pytest, GitHub Actions

Experience

Cybersecurity Research Engineer

Jan 2025 – Aug 2025

The Upanzi Network, Carnegie Mellon University Africa, Kigali, Rwanda

- Ran full-cycle penetration tests on web applications, APIs and Linux servers, including externally facing apps: testing, formal findings reports, walkthroughs with system owners, remediation tracking to closure, and retesting. Typical findings: misconfigurations, default credentials, outdated vulnerable software, IDOR.
- Designed, deployed, operated and analysed internet-facing honeypots (Cowrie, OpenCanary) for the MIMON and ATCHEDI projects over about a year. Captured malware samples, attacker commands, credentials and source data; fed indicators to ELK dashboards and MISP. Received an individual AfricaCERT Recognition and Appreciation Award (2024).
- Deobfuscated attacker commands and analysed captured malware with CyberChef, online sandboxes and Ghidra.
- Automated server hardening and rootkit scanning (applied to every new server and before each penetration test), VAPT reconnaissance, log collection and agent deployment in Python and Bash. Wrote runbooks for deployment, updates and hardening.
- Mentored three interns and student project groups.

Research Associate

Jan 2023 – Dec 2024

The Upanzi Network, Carnegie Mellon University Africa, Kigali, Rwanda

- Monitored 15+ servers across research, development and monitoring projects with Elastic Security, Security Onion and Suricata. Triaged high alert volumes, ran threat hunts, and escalated findings to project leads.
- Deployed and maintained Elastic Agent with Elastic Defend EDR and Wazuh agents. Responded to endpoint alerts, including isolating a host on a malicious-download detection and releasing it after verifying the file was an authorised security tool.
- Managed SSH keys, sudo rights and access reviews. Configured firewall rules, VPN access, network segmentation and server backups.
- Integrated MISP threat intelligence with the ELK stack and Suricata to enrich alerts.
- Co-authored a study of 224 financial Android apps from across Africa: 95% leaked secrets such as API keys, signing keys and payment-gateway credentials.
- First author of a peer-reviewed ACM paper assessing 18 African mobile banking apps against OWASP MASVS; presented it as an invited talk at the University of Washington.

Mobile Application Security Research Intern

Jun 2023 – Sep 2023

Approov Mobile Security, United Kingdom (remote)

- Tested financial Android apps against OWASP MASVS using static analysis (MobSF, jadx, apktool) and dynamic analysis (Frida, Objection, SSL-pinning bypass).
- Co-wrote an industry report on mobile app security in Africa and presented findings in live webinars, working remotely across time zones.

Cybersecurity Intern

Jun 2022 – Aug 2022

The Upanzi Network, Carnegie Mellon University Africa, Kigali, Rwanda

- Ran security scans and tests, and set up lab environments and tooling.

IT Support Specialist

Sep 2021 – Jun 2022

Carnegie Mellon University Africa, Kigali, Rwanda

- Supported 100+ staff and students across hardware, software, network and AV, and wrote troubleshooting guides.

Teaching Assistant

Mar 2019 – Jun 2021

Harare Institute of Technology, Zimbabwe

- Taught labs and tutorials for 70+ students per term and supervised student projects; two teams earned Distinctions.

ICT Officer (ICT Intern, Aug 2015 – Aug 2016)

Sep 2017 – Jul 2018

Zimasco (Pvt) Ltd, Kwekwe, Zimbabwe

- Provided Tier II IT and security support to 500+ users. Administered Active Directory, Windows Server, LAN/WAN and backups.
- Maintained endpoint antivirus protection and configured email filtering rules.
- Supported internal IT audits: gathered evidence, fixed findings and checked compliance with IT policies.

Project

SOC homelab and AI triage agent

2026

github.com/tchiboora/cybersecurity-soc-homelab | write-ups at chiboorat.medium.com

- Built a segmented attack-and-detect lab (Proxmox, OPNsense, Suricata, Wazuh, Kali). Wrote a Wazuh rule that flags a successful SSH login after repeated failures, validated against real logs and mapped to MITRE ATT&CK.
- Built an offline LLM alert-triage agent with severity computed in code, then attacked it with prompt injection and OWASP LLM test cases. Fixed the two gaps the tests exposed and added CI regression tests.

Publications and talks

- **Chiboora, T. H.**, Chacha, L., Byagutangaza, T., Gueye, A. Evaluating Mobile Banking Application Security Posture Using the OWASP's MASVS Framework. *ACM COMPASS '23*, pp. 99–106. doi.org/10.1145/3588001.3609367
- Musiime, J., Gueye, A., Atombo-Sackey, N. A., Chacha, L., **Chiboora, T.**, et al. Cookie Consent and Digital Privacy in Africa: An Analysis of Website Practices and Existing Regulatory Gaps. *ACM COMPASS 2025*.
- Byagutangaza, T., Chacha, L., **Chiboora, T. H.**, Musiime, J. J., McGregor, G. The Security Challenges of Financial Mobile Apps in Africa. *Approov / CyLab-Africa report*, 2023.
- Invited talk, University of Washington Change Seminar, February 2024.

Education

- MS Cybersecurity & Privacy, New Jersey Institute of Technology (expected Dec 2026)
- MS Information Technology, Carnegie Mellon University Africa (2023)
- BTech (Hons) Information Technology, First Class, Harare Institute of Technology (2017)

Certifications and awards

- Cisco Certified CyberOps Associate
- CompTIA PenTest+
- ISC2 Certified in Cybersecurity (CC)
- AfricaCERT Recognition and Appreciation Award, 2024

Languages: English (professional), Shona (native)